

Tio Marello

Medical Student @ Tufts School of Medicine

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Education

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| M.D. Tufts University School of Medicine

B.S. Boston College, Biology
Graduated with a Bachelor's in Biology, working in the intersection of biology, technology, and physics. <ul style="list-style-type: none"> • Worked in a wet-lab context with the aim of synthesizing nucleic acid probes for biosensors. • Created portable biosensing solutions using self-taught python and PCB design skills, allowing for the publication of two academic papers on biosensing. | Boston, MA
Jan 2026 – present

Chestnut Hill, MA
Aug 2021 – May 2025 |
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Research Experience

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| Burch Lab, Boston College, Research Assistant <ul style="list-style-type: none"> • Assist in the fabrication of graphene-based biosensors for waste-water epidemiology. • Collaborate with cross-disciplinary partners in both Biology and Engineering to create widely adoptable, scalable biosensing solutions. | Boston, MA
Nov 2022 – May 2025 |
| Meyer Lab, Boston College, Research Assistant <ul style="list-style-type: none"> • Work directly with a graduate student and assist in the development of oligonucleotide-based aptamer probes. • Create, run, and analyze PAGE and Agarose gels to quantify nucleotide synthesis efficacy. | Chestnut Hill, MA
Aug 2023 – May 2024 |

Publications

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| Electrical readout strategies of GFET biosensors for real-world requirements
Michael Geiwitz, Owen Rivers Page, Marina E Nichols, Tio Marello , Catherine Hoar, Deji Akinwande, Michelle M Meyer, Kenneth S Burch
(Biosensors and Bioelectronics) | 2026 |
| Graphene Multiplexed Sensor for Point-of-Need Viral Wastewater-Based Epidemiology
Michael Geiwitz, Owen Rivers Page, Tio Marello , Marina E. Nichols, Narendra Kumar, Stephen Hummel, Vsevolod Belosevich, Qiong Ma, Tim van Opijnen, Bruce Batten, Michelle M. Meyer, Kenneth S. Burch
10.1021/acsabm.4c00484 (ACS Applied Bio Materials) | July 2024 |

Clinical Experience

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| Boston Eye Physicians and Surgeons , Ophthalmic Technician and Scribe <ul style="list-style-type: none"> • Perform initial work-up of patients in an ophthalmic clinic, taking a ocular history and vital signs. • Act as a scribe working directly with an ophthalmologist. | Brookline, MA
May 2025 – Apr 2026 |
| Armstrong Ambulance , EMT-B <ul style="list-style-type: none"> • Administer emergency medical care in emergent situations with thorough patient assessment and clinical excellence. | Arlington, MA
June 2024 – May 2025 |

Projects

Handheld Graphene Biosensing Platform

Jan 2023 – Jan 2025

Handheld electronics platform for in-field deployment of graphene biosensors.

- Custom designed and manufactured electronics package, including custom designed PCB
- Programmed using Python and Matplotlib.
- Adopted and extensively used within the Burch Lab @ Boston College for biosensing experiments.

Selected Honors

- Inaugural Questbridge Match Scholar at the Tufts School of Medicine
- Inaugural Questbridge Match Scholar at Boston College

Volunteering Experience

Harvard Square Homeless Shelter, Volunteer, Resource Advocate, Technology Director

Cambridge, MA

- Provide space, food, and community to 16 unhoused guests across 6 months.
- Develop information management systems to further organizational goals of serving unhoused guests.

June 2023 – Aug 2025